Real-Time Multimodal Emotion Monitoring Application with Intelligent Audio-Text Fusion

Project Description

This project develops a **desktop or web-based application** that monitors and classifies human emotions in real time using both speech and text input. The system addresses the user problem of detecting emotional states during online interviews, meetings, or conversations to support emotion-aware or deception-aware analysis.

The application will include:

Core Features:

1. Live audio recording and speech-to-text transcription
2. Real-time emotion prediction (audio, text, and fused output)
3. Attention-based fusion to dynamically weight modalities
4. Visual dashboard displaying emotion timeline and confidence scores

The system will use pretrained RoBERTa and speech embeddings evaluated on IEMOCAP. A functional UI will allow users to upload recordings or perform live analysis. The final deliverable is a working multimodal emotion monitoring tool suitable for demonstration and evaluation.

Expected Outcome

The project delivers a fully functional web/desktop application that monitors and visualizes human emotions in real time using both speech and text. It features an intelligent audio-text fusion system with live analysis, emotion trend visualization, and interactive testing capabilities